

AI Resume–Job Matching with Deep Learning

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Abstract—The rapid growth of online recruitment platforms has resulted in a massive increase in the number of applications submitted for each job opening, making manual resume screening both time-consuming and inefficient. Traditional keyword-based matching approaches often fail to capture the semantic relationship between candidate qualifications and job requirements, leading to inaccurate recommendations and reduced matching performance. To address this challenge, this project proposes an artificial intelligence-based resume–job matching system using natural language processing techniques. The resume and job posting texts are first preprocessed and transformed into vector representations for similarity-based ranking. The study evaluates traditional information retrieval methods, including TF-IDF with cosine similarity and BM25, alongside transformer-based Sentence-BERT models for semantic matching. Experimental results show that the Hybrid Fine-tuned Sentence-BERT with Skill Overlap model achieved the best overall performance, obtaining a Top-1 Accuracy of 0.811 and a Mean Reciprocal Rank (MRR) of 0.849, significantly outperforming conventional keyword-based approaches.

Index Terms—TF-IDF, BM25, Natural Language Processing, Sentence-BERT, Semantic Similarity

I. INTRODUCTION

Recruitment platforms and online hiring systems generate massive numbers of applications for available positions, making candidate selection increasingly difficult for organizations. Many automated screening tools still rely on direct keyword comparison between resumes and job descriptions, which often produces unreliable matching results. Applicants with suitable qualifications may not be identified when different wording or alternative skill descriptions are used, while unrelated candidates may appear relevant because of repeated keywords. These limitations reduce the effectiveness of automated hiring systems and create a need for smarter approaches capable of understanding contextual and semantic relationships between candidate profiles and job requirements.

Recent advancements in artificial intelligence and natural language processing have introduced more intelligent approaches for understanding textual data. Deep learning models based on transformer architectures, particularly BERT and Sentence-BERT, have significantly improved the ability of machines to understand semantic meaning and contextual relationships in language. Unlike traditional keyword-based methods, these models generate dense vector representations known as embeddings, which capture the contextual meaning of entire sentences and documents. This allows semantically similar resumes and job descriptions to be matched even when they use different vocabulary or writing styles.

In this project, an intelligent resume–job matching system based on natural language processing and deep learning techniques is developed and evaluated. The proposed system preprocesses resumes and job postings by cleaning textual data, removing unnecessary symbols, and standardizing document content before transforming the text into vector representations suitable for similarity-based ranking. Multiple ranking approaches are investigated and compared in order to evaluate their effectiveness for automated recruitment systems.

The study first evaluates traditional information retrieval approaches, including TF-IDF with cosine similarity and BM25 ranking. These methods serve as baseline techniques for comparison with transformer-based semantic matching models. A pretrained Sentence-BERT model is then applied to generate semantic embeddings for resumes and job descriptions. Furthermore, the project fine-tunes Sentence-BERT using recruitment-oriented training pairs so that semantically related resume–job pairs are positioned closer together in the embedding space.

In addition to semantic similarity, this work also introduces a hybrid reranking approach that combines fine-tuned Sentence-BERT similarity with explicit skill overlap information. The motivation behind this approach is that semantic embeddings provide strong contextual understanding, while skill overlap contributes interpretable matching information related to candidate expertise and job requirements. By integrating both signals, the proposed hybrid model aims to improve ranking precision and recommendation quality.

Experimental evaluations were conducted using large-scale resume and LinkedIn job posting datasets. The performance of all methods was evaluated using ranking-based metrics such as Top- K Accuracy, Precision@ K , and Mean Reciprocal Rank (MRR). The experimental results demonstrate that transformer-based semantic matching methods outperform traditional keyword-based approaches by a significant margin. Among all evaluated methods, the Hybrid Fine-tuned Sentence-BERT with Skill Overlap model achieved the best overall performance, obtaining a Top-1 Accuracy of 0.811 and an MRR score of 0.849.

The main contributions of this project can be summarized as follows:

- Development of a complete natural language processing pipeline for automated resume–job matching.
- Comparative evaluation of classical information retrieval methods and transformer-based semantic embedding models.

- Fine-tuning of Sentence-BERT for recruitment-oriented semantic similarity learning.
- Introduction of a hybrid reranking framework that combines semantic similarity with explicit skill overlap.
- Comprehensive experimental analysis using ranking metrics and qualitative evaluation.

The remainder of this report is organized as follows. Section II defines the research problem and discusses the challenges associated with semantic resume-job matching. Section III reviews previous studies and related work in recruitment automation, information retrieval, and transformer-based semantic ranking systems. Section IV describes the datasets, preprocessing pipeline, and data preparation techniques used in this project. Section V explains the proposed methodology, including the implemented retrieval models and hybrid reranking approach. Section VI presents the experimental setup and evaluation metrics. Section VII discusses the experimental results, comparative analysis, and qualitative observations. Section VIII highlights the limitations of the current system and suggests possible future improvements. Finally, Section IX concludes the report and summarizes the main findings of the project.

Workload Distribution

The selection of the project topic and the identification of suitable datasets were carried out collaboratively by all team members. The overall implementation and report preparation tasks were then divided among the group members according to different responsibilities.

Muhammad Haseeb Muslim was primarily responsible for the initial stages of the project, including dataset collection, loading, inspection, preprocessing, and merging of the resume and job posting datasets. He also conducted exploratory data analysis to better understand the structure, distribution, and quality of the data before model development.

Mohamad Rami Karkour focused on the development of the baseline retrieval and evaluation pipeline. His responsibilities included category-to-job keyword mapping, assigning matched categories to job postings, constructing the evaluation dataset, saving processed mappings and evaluation files, and implementing the evaluation metric functions. In addition, he implemented and evaluated the TF-IDF with cosine similarity approach, BM25 ranking, reloading of processed evaluation data, and the pretrained Sentence-BERT semantic similarity model.

Sahel Nasrullah handled the deep learning and experimental stages of the project. His contributions included creating the train, validation, and test splits, generating positive resume-job training pairs, implementing and fine-tuning the Sentence-BERT model, generating training loss visualizations, saving trained models, and evaluating the fine-tuned semantic retrieval system. He also implemented the fair test-set comparison experiments, hybrid reranking strategy, and hybrid weight ablation analysis.

The final plots, result tables, qualitative analysis, and failure-case analysis were completed collaboratively during

group meetings where all members participated in the discussion and interpretation of the experimental findings.

The report writing process was also divided among the team members. Muhammad Haseeb Muslim prepared the Introduction, Problem Definition, and Related Work sections. Sahel Nasrullah wrote the Dataset Description and Preprocessing as well as the Methodology sections. Mohamad Rami Karkour was responsible for the Experimental Setup, Results and Discussion, Limitations, and Conclusion sections. The references were given by all team members.

II. PROBLEM DEFINITION

Organizations and online recruitment platforms continuously process large volumes of resumes and job advertisements across multiple professional domains. Identifying suitable candidates for specific positions has become increasingly challenging because resumes and job postings often contain unstructured text, inconsistent terminology, and varying descriptions of similar skills or experiences. As a result, accurately ranking job opportunities for candidates or matching candidates to relevant positions remains a difficult information retrieval problem.

Conventional recruitment systems primarily depend on lexical matching techniques that compare exact keywords between resumes and job descriptions. These approaches are limited because semantically related terms may be expressed differently across documents. For example, a candidate may describe experience using terms such as “software engineering” while a job posting may use phrases like “application development” or “backend programming.” Traditional keyword-focused systems frequently fail to recognize these semantic relationships, leading to inaccurate rankings and reduced recommendation quality.

Another challenge is that recruitment datasets rarely contain manually annotated relevance labels that directly indicate whether a resume is suitable for a particular job. This makes supervised training and evaluation difficult in real-world hiring scenarios. Furthermore, resumes and job postings vary significantly in structure, writing style, and detail level, increasing the complexity of semantic matching.

The main problem addressed in this project is the development of an intelligent resume-job ranking system capable of retrieving and ranking relevant job postings for a given resume using semantic understanding rather than only lexical overlap. The project also investigates whether transformer-based sentence embedding models can improve ranking quality compared to classical information retrieval methods such as TF-IDF and BM25.

In addition, the project explores whether combining semantic similarity with explicit skill-overlap information can further improve ranking accuracy and retrieval consistency. To evaluate the effectiveness of different approaches, multiple ranking models are compared using retrieval-oriented evaluation metrics including Top-K Accuracy and Mean Reciprocal Rank (MRR).

The objective of the proposed system is therefore to produce a scalable and accurate semantic retrieval framework that can better capture contextual relationships between resumes and job postings while improving recommendation quality in automated recruitment environments.

III. RELATED WORK

Automated recruitment systems have attracted significant attention in both academia and industry because of the increasing demand for scalable hiring solutions. Early resume screening systems primarily relied on keyword matching and rule-based filtering techniques. These approaches focused on identifying exact matches between candidate skills and job requirements. Although simple to implement, rule-based systems often struggle with ambiguity, synonym usage, and contextual understanding.

TF-IDF has historically been one of the most widely used text representation techniques in information retrieval systems. TF-IDF assigns weights to terms based on their frequency within a document and across a collection of documents. When combined with cosine similarity, TF-IDF can identify documents with similar keyword distributions. Several studies have applied TF-IDF to resume classification and job recommendation tasks because of its simplicity and computational efficiency. However, TF-IDF representations are sparse and fail to capture semantic relationships between words.

BM25 is another popular information retrieval method commonly used in search engines and document ranking systems. BM25 improves upon traditional TF-IDF scoring by considering document length normalization and probabilistic relevance estimation. Previous research has shown that BM25 often outperforms standard TF-IDF in retrieval-based applications. Nevertheless, BM25 remains dependent on lexical overlap and cannot effectively model contextual semantics.

The introduction of transformer-based language models significantly improved semantic representation learning. BERT demonstrated strong contextual language understanding through bidirectional transformer architectures trained on large-scale corpora. Sentence-BERT extended the BERT framework by generating semantically meaningful sentence embeddings suitable for similarity comparison using cosine distance.

Sentence-BERT has been successfully applied to semantic search, question answering, recommendation systems, and document retrieval. Its ability to generate dense semantic embeddings makes it highly suitable for resume-job matching applications. Instead of relying on exact keyword overlap, semantic embeddings capture contextual meaning and allow systems to identify conceptually related documents.

Several recent studies have explored deep learning methods for recruitment systems. Some approaches use classification-based architectures to predict candidate-job compatibility, while others rely on embedding similarity techniques. However, many existing systems depend on manually annotated relevance labels, which are difficult and expensive to obtain in recruitment domains.

Contrastive learning has emerged as an effective technique for fine-tuning semantic embedding models. MultipleNegativesRankingLoss is widely used in semantic retrieval tasks because it encourages positive document pairs to have higher similarity while implicitly treating other examples within the batch as negatives. This approach improves retrieval quality without requiring explicit negative labels.

Hybrid retrieval systems combine multiple ranking signals to improve performance. Recent research shows that combining semantic embeddings with keyword-based or skill-based features can improve retrieval robustness and interpretability. Inspired by these findings, this project integrates fine-tuned Sentence-BERT similarity with skill-overlap reranking to improve ranking quality.

Despite recent progress, automated recruitment systems still face challenges related to fairness, explainability, weak labeling, and domain adaptation. This project focuses specifically on semantic ranking quality while acknowledging broader ethical and operational considerations in real-world recruitment systems.

IV. DATASET DESCRIPTION AND PREPROCESSING

A. Datasets

Two publicly available Kaggle datasets were used in this project. The first dataset contains resume data categorized according to professional domains, while the second dataset contains LinkedIn job postings and associated skill information.

The resume dataset, as shown in Figure 1, includes resumes from multiple categories such as information technology, finance, healthcare, human resources, engineering, and sales. Each resume contains textual information describing candidate skills, education, certifications, projects, and work experience.

	resume_id	resume_text	resume_category	resume_text_clean
0	16852973	HR ADMINISTRATOR/MARKETING ASSOCIATE...	HR	hr administrator marketing associate hr admini...
1	22323967	HR SPECIALIST, US HR OPERATIONS ...	HR	hr specialist us hr operations summary versati...
2	33176873	HR DIRECTOR Summary Over 2...	HR	hr director summary over 20 years experience l...
3	27018550	HR SPECIALIST Summary Dedic...	HR	hr specialist summary dedicated driven and dyn...
4	17812897	HR MANAGER Skill Highlights ...	HR	hr manager skill highlights hr skills hr depar...

Fig. 1. Resume Dataset

The LinkedIn job posting dataset contains job titles, descriptions, required skills, and additional metadata. The dataset, as shown in figure 2, also includes mappings between jobs and skill categories, which were later used for hybrid reranking.

	job_id	title	job_skills
0	3904360847	Leasing Agent - Hendricks Commercial Properties	Management, Sales
1	3903842264	SECURITY OFFICER - STARTING WAGE \$19.00	Information Technology, Management, Other
2	3889105026	Preschool Assistant Teacher- SIGN ON BONUS	Education, Training
3	3887893610	PT Clerk - Front End - 0808	Business Development, Sales
4	3904576808	Technical Admin - Test & Validation	Administrative
5	3901952533	Sales Associate	Business Development, Sales
6	3894207430	Customer Success Manager	Other
7	3905245884	Dietitian I - Housewide Dietitians	Other
8	3902755618	ServiceNow Platform and Systems Administrator,...	Information Technology
9	3905670980	Kitchen Leader	Management, Manufacturing
10	3906230675	Systems Integration Manager	Information Technology, Project Management

Fig. 2. Job Skill Dataset

The final processed dataset contained:

- 1,935 resumes
- 6,879 job postings
- 17 valid categories

A stratified train-validation-test split was created to ensure balanced category distribution:

- Training resumes: 1,354
- Validation resumes: 194
- Test resumes: 387

B. Text Cleaning

The raw datasets contained noisy and inconsistent text formatting. Therefore, several preprocessing operations were applied before model training and evaluation.

The preprocessing pipeline included:

- Lowercase conversion
- Removal of special characters
- Removal of HTML artifacts
- Removal of unnecessary whitespace
- Token normalization
- Skill extraction and mapping

Text cleaning improves consistency and reduces irrelevant textual variations that could negatively affect semantic similarity calculations.

C. Category Mapping

The datasets did not contain explicit resume-job relevance labels. Therefore, weak supervision was used to generate relevance mappings. Resume categories were matched against job titles and skill categories using manually constructed keyword mappings.

For example, resumes labeled under the “HR” category were considered relevant to jobs containing terms such as “Human Resources”, “Recruitment”, or related HR skill mappings. Similarly, some other matched categories are shown in figure 3.

	job_id	title	job_skills	matched_categories
0	3904360847	Leasing Agent - Hendricks Commercial Properties	Management, Sales	[SALES]
1	3903842264	SECURITY OFFICER - STARTING WAGE \$19.00	Information Technology, Management, Other	[INFORMATION-TECHNOLOGY]
2	3889105026	Preschool Assistant Teacher- SIGN ON BONUS	Education, Training	[TEACHER]
3	3887893610	PT Clerk - Front End - 0808	Business Development, Sales	[BUSINESS-DEVELOPMENT, SALES]
4	3904576808	Technical Admin - Test & Validation	Administrative	[]
5	3901952533	Sales Associate	Business Development, Sales	[BUSINESS-DEVELOPMENT, SALES]
6	3894207430	Customer Success Manager	Other	[]
7	3905245884	Dietitian I - Housewide Dietitians	Other	[]
8	3902755618	ServiceNow Platform and Systems Administrator,...	Information Technology	[INFORMATION-TECHNOLOGY]
9	3905670980	Kitchen Leader	Management, Manufacturing	[]
10	3906230675	Systems Integration Manager	Information Technology, Project Management	[INFORMATION-TECHNOLOGY]

Fig. 3. Matched categories to jobs

This weak-labeling strategy enabled construction of positive resume-job training pairs for fine-tuning Sentence-BERT.

D. Skill Extraction

Skill overlap information was extracted from LinkedIn job skill mappings and processed resume text. A lightweight overlap score was calculated based on shared skills between resumes and jobs.

This skill overlap signal was later incorporated into the hybrid reranking strategy.

V. METHODOLOGY

A. System Overview

The proposed system follows a multi-stage ranking pipeline. First, resumes and job descriptions are cleaned and normalized. Next, textual representations are generated using different retrieval methods. Finally, ranking scores are computed to recommend the most relevant jobs for each resume.

The overall workflow is illustrated conceptually as follows:

- 1) Resume preprocessing
- 2) Job posting preprocessing
- 3) Vector representation generation
- 4) Similarity score computation
- 5) Ranking and reranking
- 6) Evaluation and analysis

Five different ranking approaches were implemented and compared.

B. TF-IDF with Cosine Similarity

TF-IDF represents documents using sparse term-frequency vectors weighted according to inverse document frequency. Each resume and job posting was transformed into a TF-IDF vector.

Cosine similarity was then used to measure the similarity between resume vectors and job vectors:

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{||A|| ||B||} \quad (1)$$

This method serves as a classical NLP baseline for comparison against semantic embedding models.

C. BM25 Retrieval

BM25 is a probabilistic ranking algorithm widely used in search engines and retrieval systems. It estimates document relevance based on query-term frequency and document length normalization.

The BM25 scoring function is defined as:

$$\text{BM25}(q, d) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, d)(k_1 + 1)}{f(q_i, d) + k_1(1 - b + b \frac{|d|}{\text{avgdl}})} \quad (2)$$

where:

- q represents the query
- d represents the document
- $f(q_i, d)$ is the term frequency
- avgdl is the average document length

BM25 provides stronger retrieval performance than standard TF-IDF for many lexical retrieval tasks.

D. Pretrained Sentence-BERT

Sentence-BERT is a transformer-based model designed for semantic similarity tasks. Unlike sparse keyword-based methods, Sentence-BERT generates dense vector embeddings that capture contextual semantic information.

In this project, resumes and job postings were encoded into sentence embeddings using a pretrained Sentence-BERT model. Cosine similarity between embeddings was used for ranking.

Semantic embeddings allow the system to identify conceptually similar resumes and jobs even when exact keywords differ.

E. Fine-Tuned Sentence-BERT

Although pretrained Sentence-BERT provides strong semantic representations, domain-specific fine-tuning can improve retrieval quality for recruitment tasks.

Fine-tuning was performed using positive resume-job pairs generated from category mappings. The training objective used MultipleNegativesRankingLoss, which is widely used in semantic retrieval applications.

The training process encourages semantically related resume-job pairs to have higher similarity while implicitly treating other examples within the batch as negative pairs.

The model was trained using contrastive learning techniques with mini-batch optimization.

F. Hybrid Semantic Reranking

The final proposed model combines fine-tuned Sentence-BERT similarity with skill-overlap reranking.

The hybrid score is computed as:

$$\text{HybridScore} = \alpha \cdot \text{SBERTSimilarity} + (1 - \alpha) \cdot \text{SkillOverlap} \quad (3)$$

where:

- α controls the importance of semantic similarity

- SkillOverlap measures explicit skill matching

Ablation experiments showed that the best performance was achieved using:

$$\alpha = 0.85 \quad (4)$$

This indicates that semantic similarity contributes most of the ranking signal, while skill overlap provides complementary refinement.

VI. EXPERIMENTAL SETUP

A. Training Environment

The experiments were implemented in Python using PyTorch and SentenceTransformers libraries. GPU acceleration was used during fine-tuning.

Main libraries included:

- PyTorch
- SentenceTransformers
- scikit-learn
- pandas
- NumPy
- matplotlib

B. Training Configuration

The fine-tuned Sentence-BERT model was trained using contrastive learning with MultipleNegativesRankingLoss.

Important training parameters included:

- Batch size: 16
- Optimizer: AdamW
- Learning rate scheduling
- Multiple training epochs
- Validation monitoring

C. Evaluation Metrics

Several ranking-based evaluation metrics were used.

1) *Top-K Accuracy*: Top-K Accuracy measures whether a relevant job appears within the top K ranked results.

2) *Mean Reciprocal Rank (MRR)*: MRR measures the reciprocal rank of the first relevant result:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i} \quad (5)$$

Higher MRR values indicate better ranking quality.

3) *Precision@K*: Precision@K measures the proportion of relevant results within the top K retrieved jobs.

VII. RESULTS AND DISCUSSION

A. Overall Model Comparison

The final evaluation compared all implemented methods on the same test set.

The results demonstrate substantial improvements when moving from classical lexical retrieval methods to semantic embedding approaches.

TF-IDF and BM25 achieved moderate ranking performance but struggled to capture contextual relationships between resumes and job postings. BM25 underperformed TF-IDF in

TABLE I
FINAL TEST SET PERFORMANCE COMPARISON

Method	Top-1	Top-3	Top-5	Top-10	MRR
TF-IDF + Cosine Similarity	0.576	0.682	0.713	0.780	0.646
BM25	0.499	0.636	0.698	0.762	0.593
Pretrained Sentence-BERT	0.615	0.757	0.791	0.819	0.693
Fine-tuned Sentence-BERT	0.788	0.871	0.899	0.904	0.831
Hybrid Fine-tuned SBERT + Skill Overlap	0.811	0.876	0.894	0.907	0.849

Top-1 Accuracy, suggesting that lexical relevance alone is insufficient for recruitment-oriented semantic matching.

Pretrained Sentence-BERT improved performance considerably by capturing semantic meaning beyond exact keyword overlap. However, the most significant improvement was achieved through domain-specific fine-tuning.

Fine-tuned Sentence-BERT achieved a Top-1 Accuracy of 0.788 and an MRR of 0.831, demonstrating that contrastive fine-tuning effectively adapts semantic embeddings for resume-job retrieval tasks.

The hybrid reranking model achieved the best overall performance with a Top-1 Accuracy of 0.811 and an MRR of 0.849. This confirms that combining semantic similarity with explicit skill overlap improves retrieval precision.

B. Top-1 Accuracy and MRR Analysis

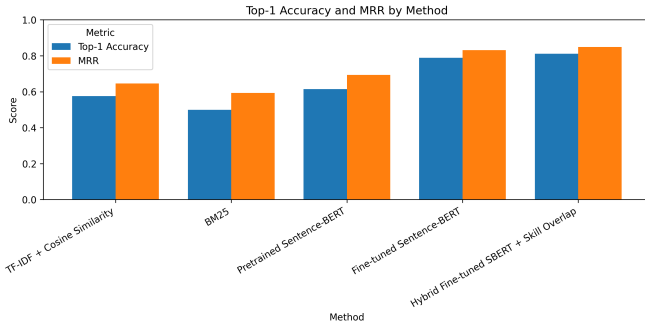


Fig. 4. Top-1 Accuracy and MRR Comparison

Figure 4 compares Top-1 Accuracy and MRR across all evaluated methods. The figure clearly illustrates the advantages of semantic embedding approaches over lexical baselines.

The hybrid model consistently achieved the strongest performance across both metrics, indicating improved ranking consistency and better placement of relevant jobs within the ranked lists.

C. Detailed Test Set Comparison

Figure 5 shows detailed Top-1, Top-3, Top-5, Top-10, and MRR results for all evaluated methods.

The performance gap between lexical retrieval and semantic retrieval becomes increasingly clear across all ranking thresholds. Fine-tuned Sentence-BERT and the hybrid model consistently outperformed classical baselines.

Interestingly, fine-tuned Sentence-BERT slightly outperformed the hybrid model on Top-5 Accuracy. However, the hybrid model achieved better Top-1 and MRR scores, suggesting that skill overlap primarily improves early ranking precision.

D. Hybrid Weight Ablation Study

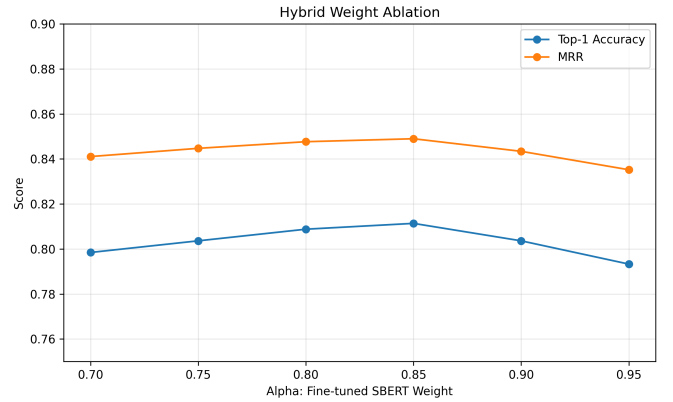


Fig. 6. Hybrid Weight Ablation Analysis

An ablation study was conducted to determine the optimal weighting between semantic similarity and skill overlap.

Figure 6 demonstrates that the best results were achieved when the semantic similarity component received approximately 85% of the total score.

This finding suggests that semantic embeddings provide the primary retrieval signal, while skill overlap contributes complementary information that improves final ranking quality.

When the skill-overlap contribution became too large, performance decreased slightly because lexical skill overlap alone cannot fully represent semantic candidate-job compatibility.

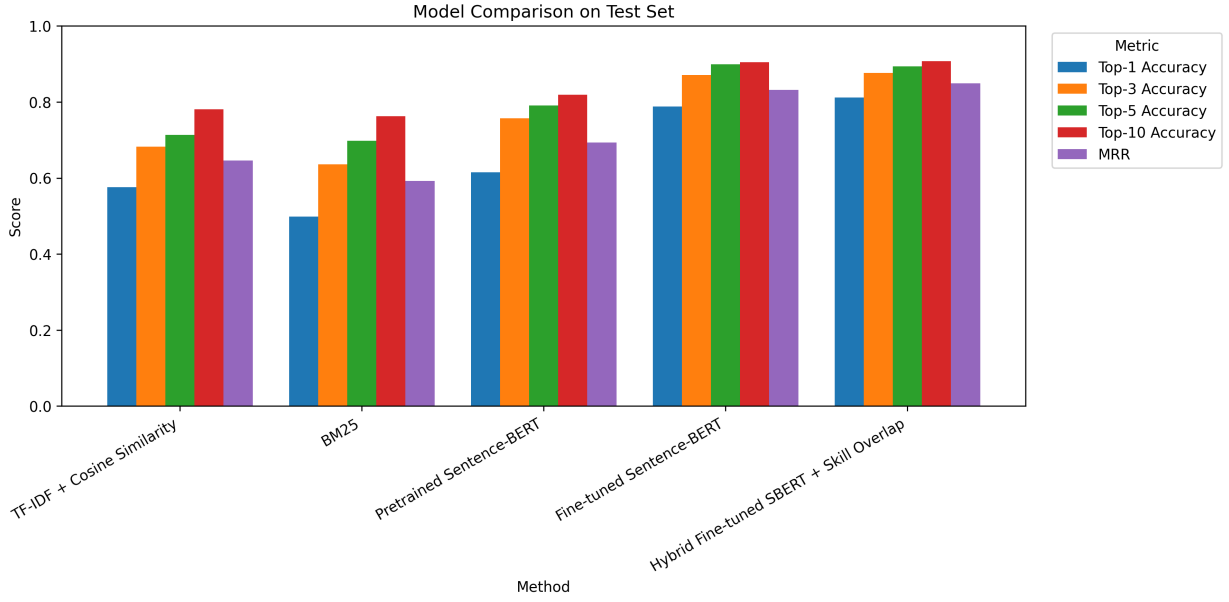


Fig. 5. Comprehensive Test Set Performance Comparison

E. Category-Wise Performance

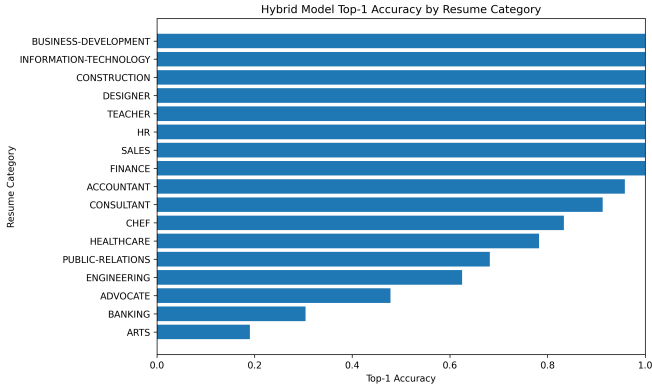


Fig. 7. Hybrid Model Top-1 Accuracy by Resume Category

Category-wise analysis revealed substantial performance differences across resume categories.

Information technology, finance, human resources, business development, and teacher categories achieved very high Top-1 Accuracy scores. These categories typically contain clearer skill terminology and more consistent semantic patterns.

In contrast, categories such as arts, banking, and advocate demonstrated lower ranking performance. These categories often involve broader or more ambiguous terminology, which increases semantic overlap between unrelated jobs.

The category-wise evaluation highlights the importance of domain-specific semantic structures in retrieval performance.

F. Training Loss Analysis



Fig. 8. Fine-Tuned SBERT Smoothed Training Loss

Figure 8 shows the training loss during Sentence-BERT fine-tuning.

The loss decreases steadily throughout training, indicating successful optimization and convergence. Although batch-level fluctuations are visible because of contrastive learning dynamics, the smoothed trend demonstrates stable learning behavior.

The reduction in training loss correlates with improved retrieval performance on the validation and test sets.

G. Qualitative Analysis

Qualitative analysis was performed to examine successful and failed retrieval examples.

The hybrid model demonstrated strong semantic understanding in many scenarios where lexical methods failed. For example, resumes describing “software development”

experience were successfully matched with jobs containing terms such as “backend engineer” and “application developer” despite limited keyword overlap.

The hybrid reranking strategy also improved retrieval precision for resumes containing explicit technical skills aligned with job requirements.

However, some failure cases were observed in categories with broad terminology or limited contextual detail. In certain situations, semantically similar but professionally unrelated jobs received high similarity scores.

These observations suggest that future systems may benefit from additional contextual metadata, hierarchical skill ontologies, or domain-specific entity extraction.

VIII. LIMITATIONS

Although the proposed system achieved strong ranking performance, several limitations remain.

First, relevance labels were generated using weak supervision based on category mapping rather than human expert annotation. While this approach enabled large-scale training and evaluation, it may introduce noise and labeling inaccuracies.

Second, some resume categories contain highly ambiguous terminology. Semantic overlap between unrelated professions can occasionally produce incorrect rankings.

Third, the skill-overlap component relies on simple lexical matching rather than advanced skill extraction models. More sophisticated entity extraction methods could improve reranking quality.

Fourth, the evaluation focused exclusively on ranking accuracy rather than fairness, bias mitigation, or ethical hiring considerations. Real-world recruitment systems must address these issues carefully.

Finally, the datasets used in this project were collected from public sources and may not fully represent real-world enterprise recruitment environments.

IX. FUTURE WORK

Several improvements can be explored in future work.

First, additional transformer architectures such as RoBERTa, DeBERTa, and instruction-tuned embedding models could be evaluated.

Second, advanced skill extraction systems using named entity recognition and ontology matching may improve hybrid reranking quality.

Third, reinforcement learning or learning-to-rank approaches could optimize ranking performance more directly.

Fourth, incorporating recruiter feedback and human annotations would enable more reliable supervised learning.

Future systems could also integrate explainability mechanisms that justify why specific candidates were matched to certain jobs.

Finally, fairness-aware ranking approaches should be investigated to reduce potential biases in automated hiring systems.

X. CONCLUSION

This project presented an AI-based resume screening and job matching system using natural language processing, information retrieval, and transformer-based semantic embeddings.

Five ranking approaches were evaluated, including TF-IDF with cosine similarity, BM25 retrieval, pretrained Sentence-BERT, fine-tuned Sentence-BERT, and a hybrid semantic reranking model.

The results demonstrated that transformer-based semantic retrieval methods substantially outperform classical lexical retrieval approaches in resume-job matching tasks.

Fine-tuned Sentence-BERT achieved strong ranking performance by learning domain-specific semantic representations using contrastive learning. Furthermore, the proposed hybrid reranking strategy improved retrieval quality by combining semantic similarity with explicit skill overlap information.

The final hybrid model achieved the best overall performance with a Top-1 Accuracy of 0.811 and an MRR of 0.849 on the test set.

The findings of this project demonstrate that semantic embedding techniques provide effective solutions for intelligent recruitment systems and automated resume screening applications.

REFERENCES

- [1] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of NAACL-HLT*, 2019.
- [2] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” *Proceedings of EMNLP*, 2019.
- [3] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [4] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing and Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [5] T. Gao, X. Yao, and D. Chen, “SimCSE: Simple Contrastive Learning of Sentence Embeddings,” *Proceedings of EMNLP*, 2021.
- [6] O. Khattab and M. Zaharia, “ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT,” *Proceedings of SIGIR*, 2020.
- [7] B. Mitra and N. Craswell, “An Introduction to Neural Information Retrieval,” *Foundations and Trends in Information Retrieval*, vol. 13, no. 1, pp. 1–126, 2018.
- [8] M. Malinowski, “AI in Recruitment and Selection: A Review of Applications and Ethical Challenges,” *International Journal of Human Resource Studies*, vol. 11, no. 2, pp. 45–60, 2021.
- [9] A. Vaswani et al., “Attention Is All You Need,” *Proceedings of NeurIPS*, 2017.
- [10] K. Henderson et al., “Efficient Natural Language Response Suggestion for Smart Reply,” *Proceedings of KDD*, 2017.